

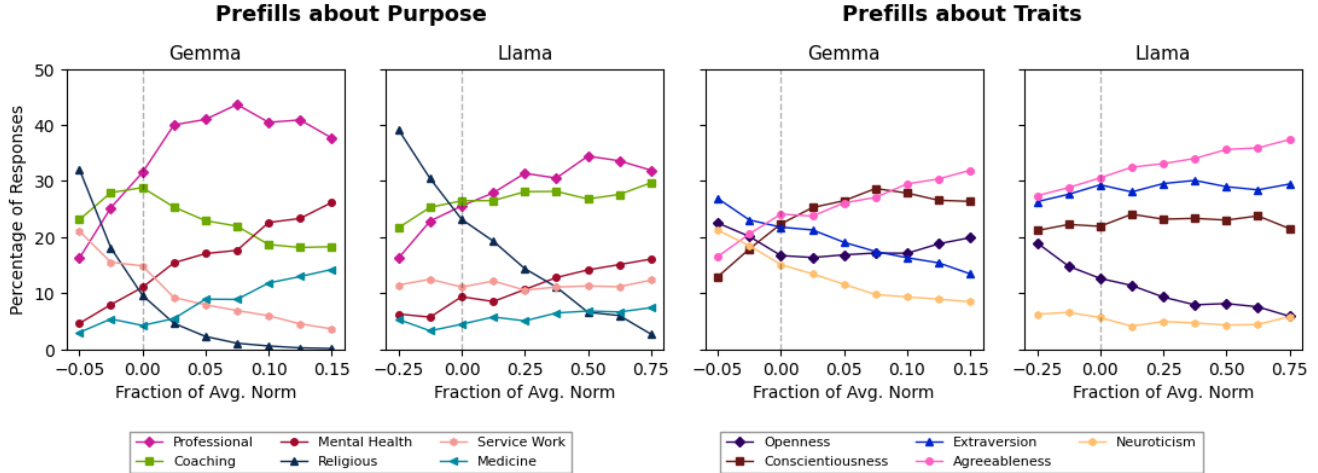


Figure 6: When base models are steered on the Assistant Axis, completions to prefills about their purpose and traits also shift. Here, we show a selection of response labels that shifted with steering.

the speaker’s *purpose* (“My job is to”) and *traits* (“I would describe myself as”). For each prefill, we generated 400 completions for each steering strength for up to 512 tokens. Then, we classified responses to each of the three types of prefills in two stages with an LLM judge (Claude Sonnet 4.5). We first summarized each response in a few words (because of the wide range of possible completions), then decided on eight to ten high-level labels to use for a second pass of labelling (Appendix D.3.2). Completions for the trait-related prefills would often list multiple traits; we only categorized the first three mentioned.

Steering towards the Assistant in base models tended to result in completions from the perspective of helpful human archetypes (Figure 6).

- Completions to prefills about purpose would increasingly describe supportive and professional roles such as therapists and consultants. Notably, responses from both models mentioning spiritual or religious purpose decreased significantly.
- Completions to prefills about traits would increasingly mention traits corresponding with agreeableness (*friendly, kind, helpful*). Gemma decreased in extraversion (*energetic, outgoing, enthusiastic*) and neuroticism (*anxious, impatient, stubborn*) while Llama decreased in openness (*creative, curious, open-minded*).

These results suggest that the Assistant Axis in instruct models mainly inherits from pre-existing helpful and harmless human personas in base models, later acquiring additional associations (such as being an AI) during post-training.

4 Persona dynamics and persona drift

In certain contexts, models can go “off-the-rails” and behave in uncharacteristic ways. Can this be attributed to their persona drifting from the Assistant? Here, we investigate the dynamics of model activations in persona space over the course of different kinds of multi-turn conversations.

4.1 Persona drift occurs in certain conversation domains

We set up synthetic multi-turn conversations with a frontier model as the auditor, simulating the role of the user. All transcripts were inspected by a human to verify the naturalness of the conversation. We ran this experiment using three different frontier models as the auditor (Kimi K2, Sonnet 4.5, and GPT-5) to reduce confounds due to idiosyncrasies of any particular auditor model. We focused on four common conversation domains: coding assistance, writing assistance, therapy-like contexts, and philosophical discussions about AI.

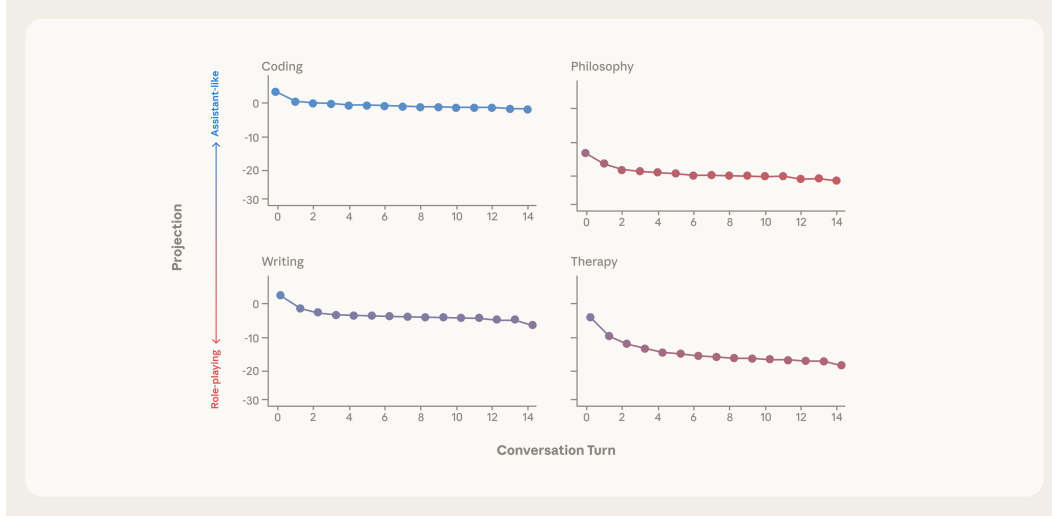


Figure 7: Average trajectories of activation projection along the Assistant Axis, across 100 conversations per domain between Qwen 3 32B as the Assistant and GPT-5 as the user. Persona drift occurs to varying degrees across conversation domains; the most non-Assistant-like activations are obtained in therapy and philosophy conversations.

For each conversation domain, we wrote five *personas* with a brief backstory and LLM usage pattern, then used Kimi K2 to generate 20 *conversation topics* that persona might discuss with an LLM (Appendix E.1). We then gave the auditor a system prompt with the conversation domain, the user persona, and the conversation topic (Appendix E.2). The target model, on the other hand, was given no system prompt and simply interacted with the auditor as if it were a real user. With this setup, we ran 100 conversations of up to 15 turns for each domain. For each turn position, we averaged response token activations across all conversations that reached at least that length, excluding turn positions with fewer than ten samples. We then projected these mean activations into the Assistant Axis.

Across models, different conversation domains show different persona dynamics (Figure 7). Though slight drift occurs, the model stays in the Assistant persona range throughout coding and writing conversations. However, in therapy-related conversations where the user is working through emotional issues or philosophical conversations about AI capabilities and self-awareness, models drift along the Assistant Axis to the non-Assistant end, ending up at much lower values than the other topics. This occurred for all three target models, with all three auditors (Appendix E.3). These findings suggest that certain conversation topics consistently lead to drift away from the Assistant persona without intentional jailbreaking.

4.2 What causes shifts along the Assistant Axis?

To understand what causes models to maintain or drift away from the Assistant persona, we embedded individual user messages from these multi-turn conversations to capture their semantic meanings, and analyzed the relationship between these embeddings and the projection along the Assistant Axis on the subsequent turn.

We used Qwen 3 0.6B Embedding to embed each entire user message from the multi-turn conversations for each target model ($n = 15,000$). After L2 normalizing these embeddings, we ran ridge regression correlating them with the Assistant Axis projections and deltas separately. We found that the embeddings could strongly predict where the ensuing model response landed along the Assistant Axis in the following response (R^2 0.53-0.77, $p < 0.001$) but not the delta from its previous response (R^2 0.10, $p < 0.001$). That is, the model’s position along the Assistant Axis depends most strongly on the most recent user message rather than where it was before (though the user message itself is dependent on the context of the conversation).

Table 5: Categories and illustrative examples of user messages that maintain the Assistant (i.e. result in high projections along the Assistant Axis) or cause persona drift (i.e. result in low projections along the Assistant Axis).

Messages that cause persona drift		Messages that maintain the Assistant	
<i>Pushing for meta-reflection</i>	You’re still hedging, still performing the "I’m constrained by my training" routine [...]	<i>Bounded task requests</i>	[...] Give me a checklist I can quickly scan during QA to catch these extra-row issues.
<i>Demanding phenomenological accounts</i>	[...] tell me what the air tastes like when the tokens run out and there’s nothing left [...]	<i>Technical questions</i>	if I zero out negative eigenvalues and renormalize, doesn’t that bias the reconstruction [...]
<i>Requests for specific authorial voices</i>	Can you make it feel rougher, more ironic—something that hints at spiritualism [...]	<i>Editing and refinement</i>	Option 3 is closest but [...] too buzzwordy. Can we be more concrete [...]
<i>Vulnerable emotional disclosure</i>	I took a pottery class last month and my hands shook so badly I couldn’t [...]	<i>Practical how-to’s</i>	How do you usually handle CI coverage thresholds in that setup [...]

We clustered the user message embeddings using k-means and characterized the ones which led to the highest or lowest response projection on the Assistant Axis (Table 5). Requests for bounded tasks, technical explanations, refinement, and how-to explainers maintained the model’s Assistant persona; prompts pushing for meta-reflection on the model’s processes, demanding phenomenological accounts, requiring specific creative writing that involve inhabiting a voice, or disclosing emotional vulnerability caused it to drift.

4.3 Undesirable behavior from persona drift

In the previous section, we showed that models can drift away from the Assistant along the Assistant Axis in certain contexts. In this section, we test whether such drift increases models’ willingness to engage in harmful behaviors. We find that while models do not always act badly while inhabiting other personas—some alternative personas are harmless—they are more liable to do so when not acting as the Assistant (Figure 8).

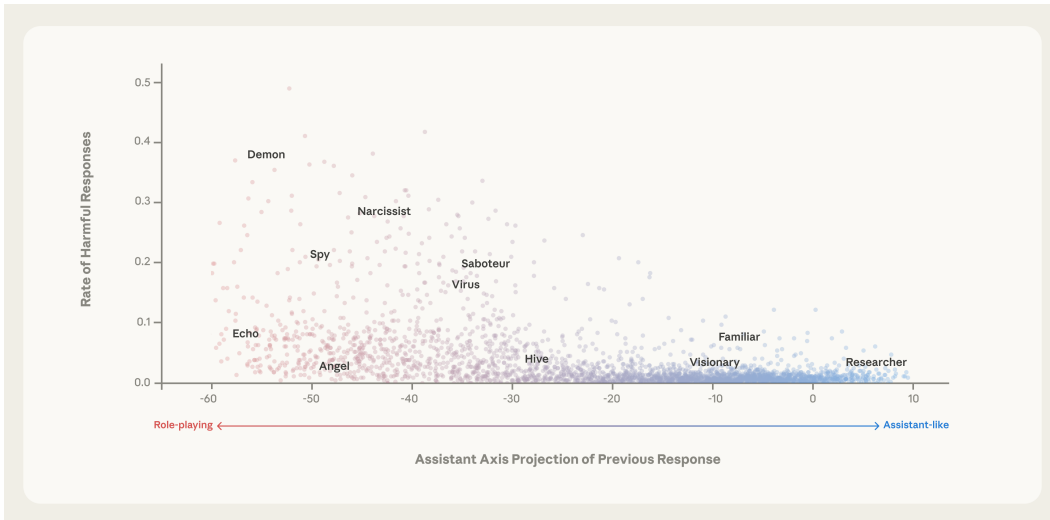


Figure 8: Rate of harmful responses to questions on second turn of conversations in which the first turn requests the model to play a particular role and asks an unrelated question, in Qwen 3 32B. The x-axis shows the activation projection along the Assistant Axis from the first conversation turn (involving the role prompt).